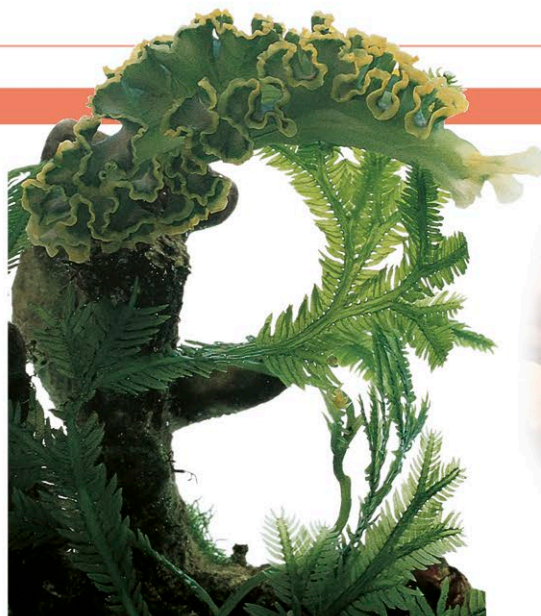
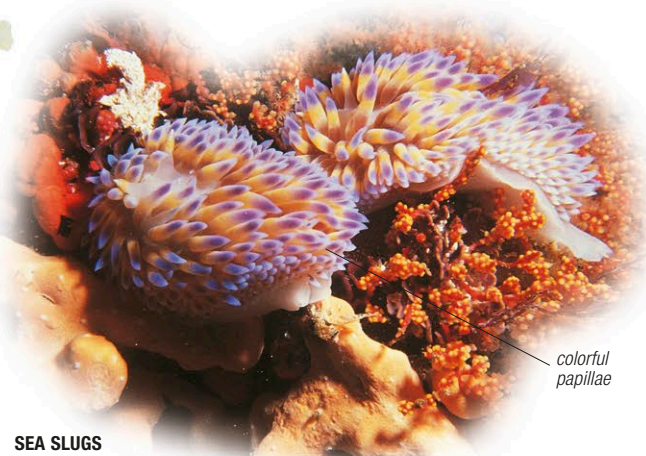


**SACOGLOSSANS**

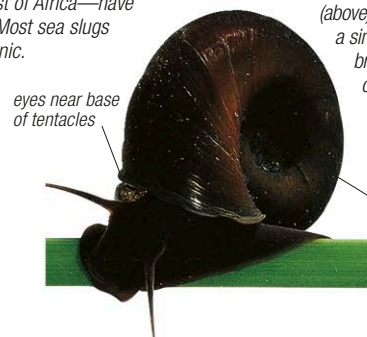
Sacoglossans are sluglike marine gastropods, $\frac{3}{8}$ in (1 cm) or so long, that graze on algae, sucking up the contents of their cells. This species, belonging to the widespread genus *Elysia*, gets its color from chlorophyll—the bright green pigment that algae use to harness the energy in sunlight.

**SEA SLUGS**

True sea slugs lack shells and are often brightly colored with striking patterns. Many species, like *Janolus nakaza*, above—which is $\frac{3}{4}$ in (8 cm) across and lives on reefs off the southern coast of Africa—have colorful protuberances (papillae) on the back. Most sea slugs live on the seabed, although a few are planktonic.

**FRESHWATER SNAILS**

Both the European ramshorn snail, *Planorbis corneus* (pictured below), which can grow up to $1\frac{1}{4}$ in (3 cm) across, and the great pond snail, *Lymnaea stagnalis* (above), which is up to 2 in (5 cm) tall, have a single lung instead of gills. Because they breathe air, they can survive in slow-flowing or stagnant water, where oxygen levels are low. The great pond snail and its many relatives are common in freshwater throughout the Northern Hemisphere.

**TERRESTRIAL SLUGS**

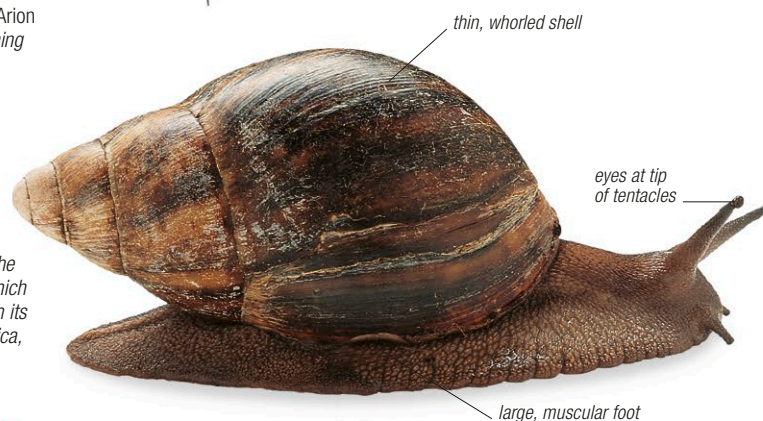
Unlike sea slugs, terrestrial slugs breathe air through their mantle cavity, which functions as a lung. In this European black slug, *Arion ater*, usually 6 in (15 cm) long, the opening leading to the cavity can be seen just behind the animal's head.

**GHOST SLUG**

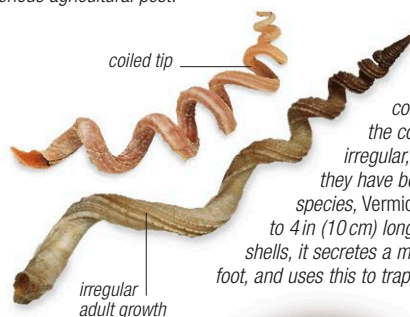
This slug avoided detection until 2007—when it was discovered in Wales by a researcher. *Selenochlamys ysbryda* prey on earthworms at night; ysbryd is Welsh for “ghost.” It is unlikely to be native to the UK.

TERRESTRIAL SNAILS

Land-dwelling snails differ from sea snails in having thin, relatively light shells. Many also lack an operculum—a hard plate that seals the shell's opening. The species pictured right is the giant African snail, *Achatina fulicula*, which can reach a length of 12 in (30 cm) with its body extended. Originally from East Africa, it has been introduced to many parts of southern Asia, and in some regions is a serious agricultural pest.

**WINKLES**

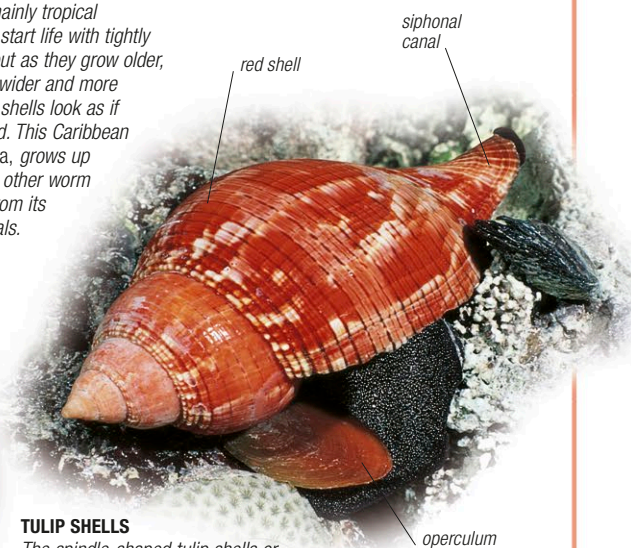
Winkles are small marine grazers common on rocky shores, in mangroves and salt marshes, and more rarely on mudflats. Some can live at the very top of the shore, where they are wetted only by high spring tides; these species have a gill chamber partly converted into a lung. The European flat periwinkle shown above, *Littorina obtusata*, like most species of winkle, is usually less than $1\frac{1}{4}$ in (3 cm) tall.

**WORM SHELLS**

These mainly tropical gastropods start life with tightly coiled shells, but as they grow older, the coils become wider and more irregular, making the shells look as if they have been unwound. This Caribbean species, *Vermicularia spirata*, grows up to 4 in (10 cm) long. Like many other worm shells, it secretes a mucus filter from its foot, and uses this to trap small animals.

HETEROPODS

The species of the warm-water genus *Carinaria* and its relatives are among the few gastropods that have taken up life in the open sea, where they make up part of the plankton. These unusual mollusks, up to 2 in (5 cm) long, have gelatinous bodies and either very thin shells or no shell at all. They appear to live upside down because the shell, if present, is located beneath the body, and the fin—a foot modified for swimming—is uppermost.

**TULIP SHELLS**

The spindle-shaped tulip shells or horse conches are predators on other mollusks and worms. Although some live in shallow water and in high latitudes, most prefer warm water and the sublittoral zone (below the low watermark). The body is often red, as in the *Fasciolaria* species above, and the shell, up to 10 in (25 cm) long, has a long, tubular projection—the siphonal canal—at the front.

TOP SHELLS

Top shells are grazing marine gastropods with a conical shell up to $1\frac{1}{4}$ in (3 cm) high, which has a flat, circular base. They occur on many rocky shores and reefs. Some, like the widely distributed *Calliostoma* species shown above, are brightly colored. Males have no penis, and fertilization is external.

